Contraction of string markings through a central factor

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Abstract

A commuting endpoint can be multiplied into the next surviving generator. The resulting tail is a central twist of the old tail; its intersection property follows directly from projection. When the lost endpoint can be recovered in the first two tail generators, the standard end criterion then gives a contraction of the entire marking. We give complete proofs, including the rank-three obstruction and the conditions for iteration. The contraction is the rank-reduction theorem of Brooksbank and Leemans; we make its common central-factor argument explicit, without claiming historical priority for the theorem or its consequences.

1 Conventions and the two elementary ingredients

A string C-marking $(s_0, \dots, s_{n-1})$ consists of distinct nonidentity involutions with $s_i s_j = s_j s_i$ when $|i - j| > 1$, and

$$\langle s_i : i \in I \rangle \cap \langle s_i : i \in J \rangle = \langle s_i : i \in I \cap J \rangle \qquad (I, J \subseteq \{0, \dots, n-1\}).$$

It is irreducible when no adjacent pair commutes. Groups need not be finite. We use the standard end criterion: a string marking is C if its two end deletions are C and their groups intersect in the middle group. This is McMullen–Schulte, Proposition 2E16, restated in [BL, Proposition 2.1]. All three premises will be checked at each use.

Lemma 1 (Forward central twist)**.** *Let $(s_0, \dots, s_{k-1})$ be a string C-marking of H, and let z be an involution in an independent direct factor. Replacing s_0 by zs_0 in $H \times \langle z \rangle$ gives a string C-marking of the subgroup generated by the new tuple. The same holds at the other endpoint.*

Proof. Projection onto H sends the new marks to the old ones, so nonidentity, distinctness and the string commutations hold. Induct on k, the rank-one case being immediate. One end deletion is the old untwisted tail. The other is a central twist of a shorter old marking, so is C by induction. An element in the intersection projects into the old middle subgroup. Since it lies in the untwisted end, it equals its projection. Every middle generator is unchanged, so the intersection is exactly the new middle subgroup. The end criterion finishes the induction. Reverse the tuple for a twist at the other endpoint. □

This is the valid forward endpoint extension of Fernandes–Leemans–Mixer, Proposition 5.3. Its converse is neither stated nor used. The same complete argument occurs in our separate work on restriction kernels; no additional result from that paper is a premise here.

Lemma 2 (The first three marks)**.** *Suppose (a, b, c) is an irreducible string C-marking. Then b does not commute with ac, and $\langle a, b, c \rangle$ is not dihedral.*

Proof. The intersection property gives $\langle a, b \rangle \cap \langle b, c \rangle = \langle b \rangle$ and makes $\langle a, c \rangle$ a Klein four group. If b commuted with ac, the central involution ac would give

$$(ab)^2 = (cb)^2 \in \langle a, b \rangle \cap \langle b, c \rangle = \langle b \rangle.$$

The square is not 1, since a, b do not commute. Nor can it be b: the equation $abab = b$ implies $aba = 1$ and hence $b = 1$. This contradiction proves the first assertion.

In a nonabelian dihedral group, every noncentral involution is a reflection. None of a, b, c can be central, because each has a noncommuting neighbour. Thus all three would be reflections. Two distinct commuting reflections in a finite dihedral group differ by its central half-turn; in an infinite dihedral group they cannot commute. In the finite case ac would therefore be central, contradicting the first assertion. To see the assertion about reflections directly, write them as $r^i t, r^j t$ with $trt = r^{-1}$. Commutativity says $r^{2(i-j)} = 1$, so their nontrivial product has order two and is central. This also proves the infinite case. □

In particular a finite dihedral group of order at least six has an irreducible C-marking precisely in rank two: a longer irreducible marking would contain a triple prohibited by Lemma 2, while the usual two reflections give rank two. This proves [BL, Lemma 2.2] without importing its cited Suzuki-group classification.

2 The contraction theorem

Theorem 3 (Brooksbank–Leemans contraction)**.** *Let $(r_0, \dots, r_{n-1})$ be an irreducible string C-marking of G, where $n \geq 4$. If*

$$r_0 \in \langle r_0 r_2, r_3 \rangle, \tag{1}$$

then

$$(r_1, r_0 r_2, r_3, \dots, r_{n-1})$$

is an irreducible string C-marking of the same group G.

Proof. The new elements are involutions. The product $r_0 r_2$ is nonidentity because the old marking has the intersection property. It cannot equal any other retained mark, since that would put an old mark in the subgroup generated by the others. The string commutations follow from the old ones. Condition (1) recovers r_0 and then r_2, so the generated group is G.

For $n = 4$ put $a = r_0, b = r_1, c = r_2, d = r_3$. Both new end pairs are C: two distinct involutions generate a rank-two C-group. Their intersection is contained in

$$\langle a, b, c \rangle \cap \langle a, c, d \rangle = \langle a, c \rangle.$$

It contains ac. If it were larger than $\langle ac \rangle$, it would be the whole Klein four group $\langle a, c \rangle$. In particular a, c would lie in $\langle b, ac \rangle$, making $\langle a, b, c \rangle = \langle b, ac \rangle$ dihedral, contrary to Lemma 2. Hence the new end intersection is exactly $\langle ac \rangle$. The end criterion proves the base case. Notice that this part proves C for the generated subgroup even without (1); that condition is needed to identify the subgroup with G.

Now let $n > 4$ and assume the result for smaller ranks. The first end of the new marking is the contraction of the old prefix through r_{n-2}. That prefix is irreducible and contains all the elements in (1), so induction makes this end C and identifies its group as

$$H = \langle r_0, \dots, r_{n-2} \rangle.$$

The other end is $(r_0 r_2, r_3, \dots, r_{n-1})$. The old intersection property and string commutations give the actual direct product

$$\langle r_0, r_2, \dots, r_{n-1} \rangle = \langle r_0 \rangle \times \langle r_2, \dots, r_{n-1} \rangle.$$

Thus Lemma 1 makes the new end C. Condition (1) identifies its group with the whole displayed product, denoted by K. The old intersection property now gives

$$H \cap K = \langle r_0, r_2, \dots, r_{n-2} \rangle = \langle r_0 r_2, r_3, \dots, r_{n-2} \rangle,$$

where the last equality again uses (1). This is precisely the new middle group, so the end criterion completes the induction.

Finally the new first adjacent pair does not commute by Lemma 2. If r_0r_2 commuted with r_3, then r_2 would commute with r_3, since r_0 does. All other adjacent pairs are unchanged. The new marking is therefore irreducible. □

3 Parity and iteration

Lemma 4. *In a string C-marking, the condition* (1) *holds if and only if r_2r_3 has finite odd order.*

Proof. The old marked subgroup is $\langle r_0, r_2, r_3\rangle = \langle r_0\rangle \times \langle r_2, r_3\rangle$. Write $m = |r_2r_3|$. If m is odd, then $((r_0r_2)r_3)^m = r_0$, proving recovery. If m is even or infinite, the dihedral group $D = \langle r_2, r_3\rangle$ has a homomorphism $\chi : D \to C_2$ taking r_2 to its nonidentity element and r_3 to 1. For finite m the defining relation $(r_2r_3)^m = 1$ respects these values exactly because m is even; for infinite m there is no such relation. The subgroup $\langle r_0r_2, r_3\rangle$ lies in the graph of χ inside $\langle r_0\rangle \times D$, and that graph excludes $(r_0, 1)$. This proves the converse. This parity argument already appears in our earlier analysis of rank-three folds; it is an explicit use of the restriction kernel, not a necessary condition for every successful contraction. □

Corollary 5. *Let the original irreducible rank-n marking have adjacent orders $p_j = |r_{j-1}r_j|$. Suppose $0 \le t \le n-3$ and $p_3, p_4, \dots, p_{t+2}$ are all finite and odd, with the condition empty when $t = 0$. Then the same group has irreducible string C-markings in every rank $n, n-1, \dots, n-t$.*

Proof. For one contraction, Lemma 4 and Theorem 3 apply. The third adjacent product in the new tuple is r_3r_4, of order p_4; subsequent adjacent products are also unchanged. Thus at the next step the same test uses p_4, then p_5, and so on. Irreducibility is preserved by the theorem at every step. Stop at rank three. This proves each indicated rank by induction. □

The first two adjacent orders can change, but neither is the next test. In particular it is unnecessary to recompute a full subgroup census after each contraction. The hypotheses and proof are those of the published rank-reduction method, here organized around the central-factor argument. No claim of historical novelty or new group existence beyond its stated applications is made.

References

[1] P. A. Brooksbank and D. Leemans, *Rank reduction of string C-group representations*, arXiv:1812.01055v2, 3 May 2019; published in Proc. Amer. Math. Soc. 147 (2019), 5421–5426.

[2] M. E. Fernandes, D. Leemans and M. Mixer, *All alternating groups A_n with $n \ge 12$ have polytopes of rank $\lfloor (n-1)/2 \rfloor$*, SIAM J. Discrete Math. 26 (2012), 482–498, Proposition 5.3.